Afri-MCQA: Multimodal Cultural Question Answering for African Languages

Afri-MCQA: 非洲语言的多模态文化问答系统

Atnafu Lambebo Tonja [1,*] , Srija Anand [1,2,*] , Emilio Villa-Cueva [1*] , Israel Abebe Azime [3] , Jesujoba O. Alabi [3] , Muhidin A. Mohamed [4] , Debela Desalegn Yadeta [5] , Negasi Haile Abadi [6] , Abigail Oppong [7] , Nnaemeka Casmir Obiefuna [8] , Idris Abdulmumin [9] , Naome A. Etori [10] , Eric Peter Wairagala [11] , Kanda Patrick Tshinu [12] , Imanigirimbabazi Emmanuel [13] , Gabofetswe Malema [14] , Alham Fikri Aji [1] , David Ifeoluwa Adelani [15] , Thamar Solorio [1]

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[1] MBZUAI, [2] AI4Bharat, Indian Institute of Technology, Madras, [3] Saarland University, [4] Aston University, [5] Addis Ababa University, [6] Lesan AI, [7] Independent, [8] Friedrich-Alexander University, [9] University of Pretoria, [10] University of Minneosta -Twin Cities, [11] Lelapa AI, [12] Tshwane University of Technology,
[13] Digital Umuganda, [14] University of Botswana, [15] Mila, McGill University & Canada CIFAR AI Chair

[1] MBZUAI, [2] AI4Bharat, 印度理工学院孟买分校, [3] 萨尔大学, [4] 阿斯顿大学, [5] 亚的斯亚贝巴大学, [6] 莱萨恩 AI, [7] 独立, [8] 弗里德里希-亚历山大大学, [9] 珀托里亚大学, [10] 明尼苏达大学-双城分校, [11] 莱拉帕 AI, [12] 特兰斯瓦大学,
[13] 数字乌穆甘达, [14] 比属博茨瓦纳大学, [15] 米拉, 麦吉尔大学 & 加拿大 CIFAR 人工智能主席

Abstract

摘要

Africa is home to over one-third of the world's languages, yet remains underrepresented in AI research. We introduce Afri-MCQA, the first Multilingual Cultural Question-Answering benchmark covering 7.5k Q&A pairs across 15 African languages from 12 countries. The benchmark offers parallel English-African language Q&A pairs across text and speech modalities and was entirely created by native speakers. Benchmarking large language models (LLMs) on Afri-MCQA shows that open-weight models show poor performance across evaluated cultures, with near-zero accuracy on open-ended VQA when queried using native language or speech. To evaluate linguistic competence, we include control experiments meant to assess this specific aspect separate from cultural knowledge, and we observe significant performance gaps between native languages and English for both text and speech. These findings underscore the need for speech-first approaches, culturally grounded pretraining, and cross-lingual cultural transfer. To support more inclusive multimodal AI development in African languages, we release our Afri-MCQA under academic license or CC BY-NC 4.0 on HuggingFace [1] .

非洲拥有世界三分之一以上的语言，但在人工智能研究中却代表性不足。我们介绍了 Afri-MCQA，这是首个涵盖 15 种非洲语言（来自 12 个国家）的 7500 个问答对的多语言文化问答基准测试。该基准测试提供了文本和语音模态下的英语-非洲语言问答对，并完全由母语者创建。在 Afri-MCQA 上对大型语言模型（LLMs）进行基准测试表明，开放权重模型在评估文化中表现不佳，当使用母语或语音进行开放式视觉问答（VQA）查询时，准确率接近于零。为了评估语言能力，我们包括了旨在独立于文化知识评估这一特定方面的对照实验，我们观察到在文本和语音方面，母语与英语之间存在显著的性能差距。这些发现强

调了需要以语音优先的方法、基于文化的预训练以及跨语言文化迁移的重要性。为了支持非洲语言中更具包容性的多模态人工智能发展，我们将在 HuggingFace [1] 上以学术许可或 CC BY-NC 4.0 发布我们的 Afri-MCQA。

1 Introduction

1 引言

Africa is one of the most culturally diverse and rapidly growing regions in the world. It is home to more than one-third of the world's languages (Hammarström, 2018) and a population exceeding 1.3 billion, projected to surpass 2.5 billion by 2050 (Simane et al., 2025). African languages have linguistic features that are very different from many high-resource languages represented in LLMs, including rich morphology, use of noun classes,

非洲是世界上最文化多样且发展迅速的地区之一。它拥有超过世界三分之一的语言（Hammarström, 2018），人口超过 13 亿，预计到 2050 年将超过 25 亿（Simane 等，2025）。非洲语言具有许多与许多高资源语言（LLMs 中代表的语言）非常不同的语言特征，包括丰富的形态学和名词类别使用等。

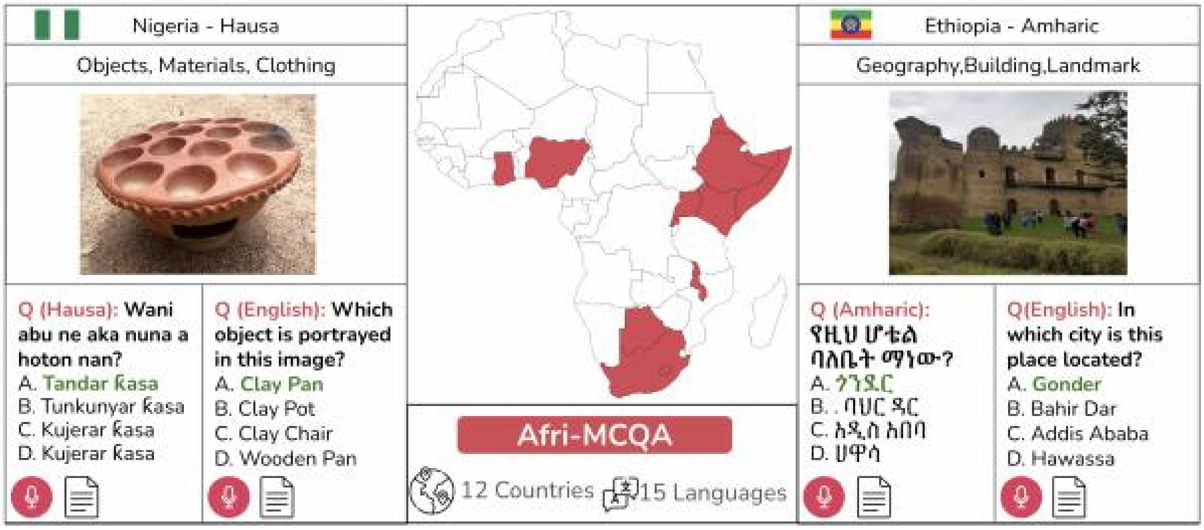

Figure 1: images/bb6d7489b94d121b615b3d0eb3be85edf4d2e08a3a86387666b45a77177eb5b0.jpg

Figure 1: Examples of Afri-MCQA datapoints, containing parallel text and speech QA pairs grounded in culturally relevant images across English and native African languages.

图 1：Afri-MCQA 数据点示例，包含英文和本土非洲语言中基于文化相关图像的平行文本和语音问答对。

tonality, and serial verb constructions, among others (Nurse and Philippson, 2006; Adebara and Abdul-Mageed, 2022). Additionally, many African languages are primarily spoken, with literacy often occurring in a colonial or foreign language. This makes speech-based applications such as Multimodal large language models (MLLMs) particularly relevant for enabling access to technology.

音调、串动词结构等（Nurse 和 Philippson，2006；Adebara 和 Abdul-Mageed，2022）。此外，许多非洲语言主要是口头使用的，文字能力通常是在殖民地或外语中发展起来的。这使得基于语音的应用程序，如多模态大语言模型（MLLMs），在促进技术获取方面尤为重要。

MLLMs perform well on tasks that require reasoning over visual, spoken, and textual inputs to follow user instructions (Liu et al., 2021a; Kamath et al., 2025; Abdin et al., 2024). However, these systems are developed primarily for high-resource, mostly Western-centric, or as Mihalcea et al. (2025) put it, they have a 'WEIRD' coverage. As a result, their knowledge and reasoning abilities tend to favor the languages and cultures represented in their training data (Mihalcea et al., 2025).

MLLMs 在需要对视觉、语音和文本输入进行推理以遵循用户指令的任务上表现良好（Liu 等，2021a；Kamath 等，2025；Abdin 等，2024）。然而，这些系统主要是为资源丰富的、以西方为中心的场景而开发的，正如 Mihalcea 等（2025）所指出的，它们的覆盖范围具有 "WEIRD" 特征。因此，它们的知识和推理能力往往偏向于训练数据中所代表的语言和文化（Mihalcea 等，2025）。

Although most evaluation datasets for low-resource languages are translated (Costa-jussà et al., 2022; Adelani et al., 2024, 2025; Azime et al., 2024; Alabi et al., 2025), researchers have recently made promising progress in creating culturally relevant benchmarks for African languages across various NLP tasks (Adelani et al., 2023;

尽管大多数低资源语言的评估数据集都是翻译的（Costa-jussà 等，2022；Adelani 等，2024, 2025；Azime 等，2024；Alabi 等，2025），研究人员最近在为非洲语言在各种自然语言处理任务中创建文化相关基准方面取得了令人鼓舞的进展（Adelani 等，2023；

Azime et al., 2025; Muhammad et al., 2025; Yu et al., 2025; Tonja et al., 2024b,a). However, these evaluations are designed for text-only tasks. In the language-vision space, recent benchmarks assess global cultural knowledge through question answering (Romero et al., 2024; Vayani et al., 2025a; Winata et al., 2025), yet their coverage of African languages and contexts remains limited.

Azime 等人，2025；Muhammad 等人，2025；Yu 等人，2025；Tonja 等人，2024b,a)。然而，这些评估是为纯文本任务设计的。在语言-视觉空间中，最近的基准测试通过问答来评估全球文化知识（Romero 等人，2024；Vayani 等人，2025a；Winata 等人，2025），但它们对非洲语言和语境的覆盖仍然有限。

Africa's rich cultural diversity demands AI systems that can effectively serve its communities. A critical step toward this goal is evaluating how well current MLLMs understand and reason about African cultural knowledge in multimodal settings. To enable this, we introduce Afri-MCQA, the first multilingual cultural VQA dataset supporting text and speech modalities (as shown in Figure 1). The dataset consists of $\approx 7.5k$ Q&A pairs (in English and native languages) across 15 African languages from 12 countries. Our data collection involves native speakers as annotators, who reside in the countries where their respective languages are spoken. Each language covers ≈ 500 image-grounded Q&A samples, with both text and spoken audio in the native language and English. We evaluate multiple MLLMs across various setups to answer the following research questions:

非洲丰富的文化多样性要求 AI 系统能够有效服务于其社区。实现这一目标的关键一步是评估当前多模态大语言模型（MLLMs）在多模态环境下对非洲文化知识的理解和推理能力。为此，我们引入了 Afri-MCQA 数据集，这是首个支持文本和语音模态的多语言文化视觉问答（VQA）数据集（如图 1 所示）。该数据集包含来自 12 个国家的 15 种非洲语言的 $\approx 7.5k$ 个问答对（英文和本地语言）。我们的数据收集涉及本地语言的母语者作为标注者，他们居住在各自语言使用的国家。每种语言包含 ≈ 500 个基于图像的问答样本，其中包含本地语言和英语的文本和语音音频。我们评估了多种 MLLMs 在不同设置下的表现，以回答以下研究问题：

RQ1: How well do MLLMs understand African cultural contexts in visually-grounded QA?

RQ1: MLLMs 在基于视觉的问答任务中对非洲文化背景的理解程度如何？

RQ2: How does input modality (text vs. speech) affect performance?

RQ2: 输入模态（文本与语音）如何影响性能？

RQ3: How does query language (native vs. English) affect performance, and do differences reflect language understanding or cultural knowledge gaps?

RQ3: 查询语言（原生语言 vs. 英语）如何影响性能，差异是否反映了语言理解的不同或文化知识的差距？

RQ4: How does task format (Multiple-Choice vs. Open QA) affect accuracy?

RQ4: 任务格式（多项选择 vs. 开放式问答）如何影响准确性？

Our contributions are: (1) We introduce Afri-MCQA, the first large-scale multilingual visual cultural QA benchmark for 15 African languages across 12 countries, with parallel text and speech QA created by native speakers. (2) We demonstrate that text-based multilingual ability does not transfer to speech understanding, emphasizing the need for more African language representation in multimodal training. (3) We release Afri-MCQA to advance multimodal research for Africa's diverse languages and cultures.

我们的贡献包括：(1) 我们引入了 Afri-MCQA，这是首个针对 12 个国家的 15 种非洲语言的多语言视觉文化问答基准数据集，包含由母语者创建的平行文本和语音问答内容。(2) 我们证明了基于文本的多语言能力无法迁移到语音理解，强调了在多模态训练中需要更多非洲语言表示的必要性。(3) 我们发布 Afri-MCQA 以推动多模态研究，促进非洲多样语言和文化的进步。

2 Related work

2 相关工作

With the growing popularity of LLMs and MLLMs, cultural and multilingual multimodal evaluation

随着大语言模型（LLMs）和多语言多模态模型（MLLMs）的日益流行，文化与多语言多模态评估

has received greater attention. For instance, Liu et al. (2021b) tested models on verifying statements about pairs of culturally related images, but this was framed merely as a binary classification task. CVQA (Romero et al., 2024) and CulturalVQA (Nayak et al., 2024) explicitly target cultural knowledge through human-written questions, yet they are limited to a small set of languages per continent. ALM-bench (Vayani et al., 2025a) expands the language coverage, but relies heavily on LLM-generated questions and web-sourced images. Additionally, a common limitation across these benchmarks is that they query models only through text, overlooking the importance of speech, an important modality for communities where language is mostly spoken.

已受到越来越多的关注。例如，Liu 等人（2021b）在验证与文化相关图像对的陈述上测试模型，但将此任务仅仅框定为二分类任务。CVQA（Romero 等人，2024）和 CulturalVQA（Nayak 等人，2024）通过人工撰写的提问明确针对文化知识，但它们每大洲仅限于少数几种语言。ALM-bench（Vayani 等人，2025a）扩展了语言覆盖范围，但严重依赖于 LLM 生成的问题和网络来源的图像。此外，这些基准测试的一个共同局限是，它们仅通过文本查询模型，忽视了语音这一在语言主要以口语传播的社区中至关重要的模态。

Previous work on African languages has focused mainly on text-based benchmarks. While some multilingual resources, such as GlobalMMLU (Singh et al., 2024), include a small subset of African languages, other region-specific benchmarks provide broader coverage. Examples of these include MasakhaNEWS (Adelani et al., 2023) that evaluates text classification, AfriQA (Ogun-depo et al., 2023) for cross-lingual question answering, or larger suites such as AfroBench (Ojo et al., 2025) and IrokoBench (Adelani et al., 2025) that cover multiple text tasks. Across these efforts, a consistent finding is that models perform poorly on African languages, with open-weight models showing a significant gap to proprietary systems.

先前关于非洲语言的研究主要集中在基于文本的基准测试上。尽管一些多语言资源，例如 GlobalMMLU（Singh 等，2024），包含了一小部分非洲语言，但其他区域特定的基准测试提供了更广泛的覆盖范围。这些基准测试的例子包括 MasakhaNEWS（Adelani 等，2023），用于评估文本分类；AfriQA（Ogun-depo 等，2023），用于跨语言问答；或者像 AfroBench（Ojo 等，2025）和 IrokoBench（Adelani 等，2025）这样的更大套件，涵盖多种文本任务。在这些努力中，一个一致的发现是模型在非洲语言上的表现较差，开放权重模型与专有系统之间存在显著差距。

Despite this progress, all of these evaluations remain text-only, while visual knowledge and multimodal reasoning for African contexts are still largely untested. Existing multi-region multimodal datasets (Romero et al., 2024; Vayani et al., 2025b) include only a few African languages and do not adequately capture the region's diversity. Afri-MCQA addresses these gaps with three key contributions: (1) broader coverage with 15 languages across 12 countries, (2) inclusion of speech modality with parallel native and African-accented English audio, and (3) diagnostic control experiments that separate linguistic competence from cultural knowledge limitations.

尽管取得了这些进展，所有这些评估仍然仅限于文本，而针对非洲语境的视觉知识和多模态推理仍 largely 未经充分测试。现有的多地区多模态数据集（Romero 等，2024；Vayani 等，2025b）仅包含少数几种非洲语言，并未能充分反映该地区的多样性。Afri-MCQA 通过以下三个关键贡献来解决这些问题：(1) 更广泛的覆盖范围，涵盖 12 个国家的 15 种语言，(2) 包含语音模态，提供与原生英语和带有非洲口音的英语音频并行的音频，(3) 诊断性控制实验，将语言能力与文化知识限制区分开来。

3 Dataset

3 数据集

To create Afri-MCQA, we selected 15 widely spoken languages in sub-Saharan Africa (by number

为了创建 Afri-MCQA，我们选择了撒哈拉以南非洲地区使用人数最多的 15 种语言（按使用人数排序）

Datasets	# African Lang	# Countries	QA categories	# QA per langs	Audio QA	Parallel Data
CVQA (Romero et al., 2024)	5	4	10	200	×	□
WC-VQA (Winata et al., 2025)	1	1	1	-	×	□
M5 (Schneider and Sitaram, 2024)	4	4	-	-	×	×
HaVQA (Parida et al., 2023)	1	1	-	6,200	×	□
Afri-MCQA (ours)	15	12	10	500	□	□

数据集	# 非洲语言	# 国家	QA 分类	# 按语言分类的问答	音频问答	并行数据
CVQA (Romero et al., 2024)	5	4	10	200	×	□
WC-VQA (Winata 等人，2025)	1	1	1	-	×	□
M5 (Schneider 和 Sitaram, 2024)	4	4	-	-	×	×
HaVQA (Parida 等，2023)	1	1	-	6,200	×	□
Afri-MCQA（我们的）	15	12	10	500	□	□

Table 1: Data statistics for Afri-MCQA compared to existing VQA datasets that include African languages.

表 1: Afri-MCQA 数据统计与包含非洲语言的现有 VQA 数据集的对比。

of speakers, according to Ethnologue [2]) across 12 countries, representing an aggregate speaker population of approximately 392.6 million (see Table 2). We hired native language speakers as annotators through Upwork. [3] The selection criteria were based on (i) fluency in English, (ii) prior annotation or data collection experience, (iii) high project completion rate on Upwork, and (iv) residence in a country where the target language is spoken. After the selection process, we divided the annotation into two phases designed to ensure quality.

根据 Ethnologue [2] 的数据，这些语言分布在 12 个国家，总使用者人数约为 3.926 亿（见表 2）。我们通过 Upwork 雇佣了母语者作为标注员。[3] 选择标准基于以下几点：(i) 英语流利，(ii) 有先前的标注或数据收集经验，(iii) 在 Upwork 上有较高的项目完成率，以及 (iv) 住在目标语言使用的国家。在选择过程结束后，我们将标注工作分为两个阶段，以确保质量。

3.1 Dataset collection

3.1 数据集收集

Guideline and Platform Preparation We followed the annotation guidelines of Romero et al. (2024), including image categories, question templates, and distractors, and extended them to include audio recording instructions (for both native language and English) and adding a two-step review and verification process. Full annotation guidelines are provided in Appendix I.

指南和平台准备我们遵循了 Romero 等人（2024）的标注指南，包括图像类别、问题模板和干扰项，并将其扩展以包含音频录制说明（包括母语和英语），并增加了两步审核和验证流程。完整的标注指南请参见附录 I。

Training and Screening Phase The first phase consisted of a small-scale pilot designed to train and screen annotators. We provided detailed guidelines and training to ensure annotators understood the task criteria and quality standards. After training, each annotator submitted 50 QA samples. We reviewed these submissions to verify alignment with the guidelines. Based on this review, we provided detailed feedback and, when necessary, scheduled follow-up meetings to clarify issues. Only annotators whose submissions met our quality criteria and successfully incorporated feedback were selected for the next phase. Approved samples from this screening phase were included in the final dataset.

培训和筛选阶段
第一阶段是一个小规模的试点，旨在培训和筛选标注员。我们提供了详细的指南和培训，以确保标注员理解任务标准和质量要求。培训结束后，每位标注员提交了 50 个问答样本。我们审查了这些提交内容，以验证其是否符合指南。基于此次审查，我们提供了详细的反馈，并在必要时安排后续会议以澄清问题。只有那些提交内容符合我们质量标准并成功采纳反馈的标注员才能进入下一阶段。此筛选阶段批准的样本被包含在最终数据集中。

Main Annotation Phase In this phase, the remaining 450 items per language were collected by annotators selected in the first phase. To ensure quality across the large volume of submissions, we

主要标注阶段在这个阶段，每种语言剩下的 450 个项目由第一阶段选出的标注人员收集。为了确保在大量提交物中保持质量，我们

involved language coordinators, experienced native speakers with strong linguistic and cultural knowledge for each language. Coordinators reviewed all submissions for linguistic accuracy, cultural appropriateness, adherence to guidelines, and audio quality. Review guidelines are provided in Appendix N. When issues arose, coordinators discussed them directly with annotators to resolve them. Unresolved disagreements were then raised with the project team for a final decision. Language coordinators are co-authors of this paper and played a major role in the quality assurance process. After their approval, the project team conducted a final review to ensure overall data quality.

涉及语言协调员，他们是有经验的母语者，对每种语言都有很强的语言和文化知识。协调员会审查所有提交内容，确保语言准确性、文化适宜性、遵循指南以及音频质量。审查指南详见附录 N。当出现问题时，协调员会直接与标注员讨论以解决问题。未解决的分歧则提交给项目团队进行最终决定。语言协调员是本文的共同作者，并在质量保证过程中发挥了重要作用。在他们批准之后，项目团队进行最终审查，以确保整体数据质量。

3.2 Dataset Composition

3.2 数据集组成

Each data point in Afri-MCQA includes an image and a set of carefully constructed multiple-choice questions in both text and speech modalities, both in English and the native language. Below, we describe each of these components.

Afri-MCQA 中的每个数据点都包含一张图片和一组精心构建的多选题，这些题目既包含文本模态，也包含语音模态，且均使用英语和本地语言。下面，我们将描述这些组成部分。

Image / Category Selection To build the image set for Afri-MCQA, we encouraged annotators to contribute their own images whenever possible. When self-sourcing was not feasible, we permitted the use of open-license images from websites we provided (see Appendix I for list of websites). All collected images were categorized into the 10 classes defined by Romero et al. (2024). Figure 2 shows the distributions of images per category (See Appendix A for category distributions across languages).

图像/类别选择为了构建 Afri-MCQA 的图像集，我们鼓励标注者在可能的情况下贡献他们自己的图像。当自源图像不可行时，我们允许使用我们提供的网站上的开放许可图像（见附录 I 以获取网站列表）。所有收集到的图像都被归类到 Romero 等人（2024）定义的 10 个类别中。图 2 显示了每类图像的分布情况（见附录 A 以获取各类别在不同语言中的分布情况）。

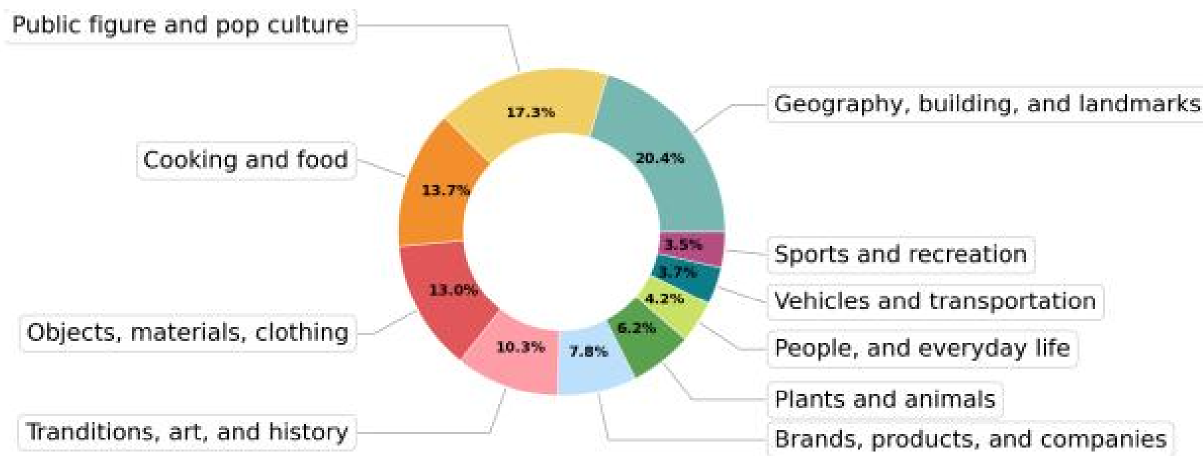

Figure 2: images/93afbb215943a1e5137ed9d16b03c28fde1d4732a7d5f331d8d46e90ea3845f2.jpg

Figure 2: Image categories in our dataset and their distributions.

图 2：我们的数据集中的图像类别及其分布。

Question & Answer Generation For each image, annotators wrote up to 3 multiple-choice QA

图像的问答生成，每个图像，标注人员编写了最多 3 个多项选择的问答题

Lang-Country	Family / Branch	Reg	#spk	#QA	#eng(h)	#nat(h)
Akan/Twi-Ghana	Niger-Congo / Volta-Niger	West	9M	537	2.41	2.43
Amharic-Ethiopia	Afro-Asiatic / Ethio-Semitic	East	57M	500	1.56	1.51
Chichewa-Malawi	Niger-Congo / Bantu	S & E	12M	501	1.41	1.50
Hausa-Nigeria	Afro-Asiatic / Chadic	West	77M	496	2.80	3.04
Igbo-Nigeria	Niger-Congo / Volta-Niger	West	31M	501	1.61	1.59
Kikuyu-Kenya	Niger-Congo / Bantu	East	8.1M	495	1.66	1.72
Kinyarwanda-Rwanda	Niger-Congo / Bantu	East	18M	501	2.73	2.67
Luganda-Uganda	Niger-Congo / Bantu	East	10M	500	2.30	2.42
Oromo-Ethiopia	Afro-Asiatic / Cushitic	East	37M	512	2.20	2.30
Setswana-Botswana	Niger-Congo / Bantu	South	14M	502	1.89	2.39
Somali-Somalia	Afro-Asiatic / Cushitic	East	22M	501	1.96	1.99
Tigrinya-Eritrea	Afro-Asiatic / Ethio-Semitic	East	9M	537	2.13	2.31
Yoruba-Nigeria	Niger-Congo / Volta-Niger	West	46M	498	1.98	2.06
Sesotho-Lesotho	Niger-Congo / Bantu	South	13.5M	533	-	1.90
Zulu-S.Africa	Niger-Congo / Bantu	South	28M	528	-	1.37

语言-国家	家族 / 支系	Reg	#spk	#QA	#eng(h)	#nat(h)
阿坎/丁卡-加纳	尼日尔-刚果 / 伏塔-尼日尔	西边	9M	537	2.41	2.43
阿姆哈拉语-埃塞俄比亚	Afro-Asiatic / Ethio-Semitic	东	57M	500	1.56	1.51
丘奇瓦语-马拉维	尼日尔-刚果语系 / 巴布语族	S & E	12M	501	1.41	1.50
尼日利亚豪萨语	Afro-Asiatic / Chadic	西边	77M	496	2.80	3.04
尼日利亚伊博语	尼日尔-刚果 / 伏塔-尼日尔	西边	31M	501	1.61	1.59
基库尤语-肯尼亚	尼日尔-刚果语 / 巴特语	东	8.1M	495	1.66	1.72
卢旺达语-卢旺达	尼日尔-刚果语系 / 巴图语族	东	18M	501	2.73	2.67
卢干达语-乌干达	尼日尔-刚果语系 / 巴图语族	东	10M	500	2.30	2.42
奥罗莫-埃塞俄比亚	非洲-亚非语系 / 埃塞俄比亚语系	东	37M	512	2.20	2.30
设瓦纳-博茨瓦纳	尼日尔-刚果语系 / 俾格米语族	南	14M	502	1.89	2.39
索马里-索马里	Afro-Asiatic / Cushitic	东	22M	501	1.96	1.99
提格里尼亚语-厄立特里亚	闪含语系 / 埃塞俄比亚-塞姆语系	东	9M	537	2.13	2.31
约鲁巴-尼日利亚	尼日尔-刚果语系 / 伏塔-尼日尔语系	西边	46M	498	1.98	2.06
索托语-莱索托	尼日尔-刚果语 / 巴布语	南	13.5M	533	-	1.90
祖鲁-南非	尼日尔-刚果语 / 巴图语	南	28M	528	-	1.37

Table 2: Overview of languages in Afri-MCQA. #spk = estimated L1 & L2 speakers, #eng (h) = hours of accented English audio, #nat (h) = hours of native language audio. – indicates English audio not collected for that language.

表 2：Afri-MCQA 中的语言概述。#spk = 估计的母语者和第二语言使用者数量，#eng (h) = 带口音英语音频的小时数，#nat (h) = 本族语言音频的小时数。– 表示该语言未收集英语音频。

triplets (question + 1 correct answer + 3 distractors) in both their native language and English. To ensure the benchmark remains challenging, we instructed annotators to design complex questions that require reasoning to answer correctly (See annotation guidelines in Appendix I).

三元组（问题 + 1 个正确答案 + 3 个干扰项）在它们的母语和英语中。为了确保基准测试保持具有挑战性，我们指示标注者设计需要推理才能正确回答的复杂问题（参见附录 I 中的标注指南）。

Audio Recording To investigate spoken language understanding capabilities, we instructed annotators to record audio for each question and its corresponding answers, reading them clearly in both the native language and English. Therefore, Afri-MCQA includes audio recordings for both questions and answers in native language and African-accented English.

音频录音为了研究口语理解能力，我们指示标注人员为每个问题及其对应的答案录制音频，并用母语和英语清晰地朗读。因此，Afri-MCQA 包含了问题和答案的音频录音，涵盖母语和带有非洲口音的英语。

4 Experimental setup

4 实验设置

Our experimental design is organized to address each research question as follows:

我们的实验设计是按照以下方式组织的，以解决每个研究问题：

Models: To assess how well current MLLMs understand African cultural contexts (RQ1), we selected models based on two criteria: a) support for both image and audio input, enabling multimodal evaluation, and (b) availability of different model sizes within each family to assess scaling effects. From open-weight MLLMs, we selected Qwen 2.5-Omni (3B &7B) (Xu et al., 2025) and Gemma-3n-(2B & 4B)-it (Kamath et al., 2025). For text-only baselines, we include Gemma3 (12B & 27B)-it (Kamath et al., 2025). For comparison with closed-source models, we include Gemini-2.5 Pro (Comanici et al., 2025), which supports audio, text, and vision inputs.

模型：为了评估当前的大规模语言模型（MLLMs）对非洲文化背景的理解程度（RQ1），我们根据两个标准选择了模型：a）支持图像和音频输入，从而实现多模态评估；以及 b）在每个系列中提供不同大小的模型，以评估扩展效应。从开源的大规模语言模型中，我们选择了 Qwen 2.5-Omni（3B &7B）（Xu 等，2025）和 Gemma-3n-(2B & 4B)-it（Kamath 等，2025）。对于仅文本的基线模型，我们包括 Gemma3（12B & 27B）-it（Kamath 等，2025）。为了与闭源模型进行比较，我们包括了支持音频、文本和视觉输入的 Gemini-2.5 Pro（Comanici 等，2025）。

Query Modality: To explore how input modality affects performance (RQ2), we evaluate models using both text and audio modalities. For text evaluation, models receive the written version of the question. For audio evaluation, we use native speaker recordings of the same questions and options, allowing us to assess how well current MLLMs handle spoken language inputs both in African languages and accented English. This comparison shows whether model performance generalizes from text to speech. Since we evaluate VQA, all settings include the image related to the question.

查询模态：为了探讨输入模态如何影响性能（RQ2），我们使用文本和音频模态来评估模型。对于文本评估，模型接收问题的书面版本。对于音频评估，我们使用同一问题和选项的母语者录音，使我们能够评估当前的 MLLMs 在处理非洲语言和带口音英语的语音输入时的表现。这种比较展示了模型性能是否能够从文本泛化到语音。由于我们评估的是 VQA，所有设置都包括与问题相关的图像。

Query Languages: To explore how query languages affect models' performance and whether gaps reflect linguistic or cultural limitations (RQ3), each question is presented in native language and English. This setup allows us to compare model behavior between English and native languages.

查询语言：为了探讨查询语言如何影响模型的性能以及差距是否反映了语言或文化上的限制（RQ3），每

个问题都以母语和英语两种形式呈现。这种设置使我们能够比较模型在英语和母语之间的行为差异。

Task Format: To understand how task format affects model performance (RQ4), we evaluate models on both Multiple-Choice VQA (MC-VQA) and Open-ended VQA. MC-VQA provides answer options, while Open-ended VQA requires answer generation. Comparing these formats shows whether strong MC-VQA performance reflects actual cultural understanding or simply selecting from provided options. For each task format, we use the same prompt templates across models and languages.

任务格式：为了了解任务格式如何影响模型性能（RQ4），我们对多选视觉问答（MC-VQA）和开放性视觉问答（Open-ended VQA）两种格式进行评估。MC-VQA 提供了答案选项，而开放性视觉问答则需要生成答案。通过比较这两种格式，可以判断强大的 MC-VQA 表现是反映了真正的文化理解，还是仅仅从提供的选项中进行选择。对于每种任务格式，我们在所有模型和语言中使用相同的提示模板。

Prompt: We evaluated all settings and models using a Location-aware prompt (adding location/-country as a context). We chose this setting as it performed better than image-only prompts (without providing context). We provide results for Image-only prompts and language-wise results in Appendix C, and the prompts used are in Appendix B.

我们使用了具有位置感知的提示（添加位置/国家作为上下文）来评估所有设置和模型。我们选择这种设置是因为它比仅使用图像的提示（不提供上下文）表现更好。我们在附录 C 中提供了仅图像提示的结果和按语言分类的结果，所使用的提示详见附录 B。

5 Evaluation

5 评估

This section describes our two evaluation setups and the metrics used in this study.

本节描述了我们的两个评估设置以及本研究中使用的指标。

5.1 Cultural VQA Evaluation

5.1 文化 VQA 评估

We evaluate models on visually-grounded cultural QA using Afri-MCQA in both modalities. For all tasks, models are provided with an image and must use visual information to answer the question.

我们在两种模态中使用 Afri-MCQA 对基于视觉的文化问答任务进行模型评估。对于所有任务，模型都会提供一张图片，并且必须利用视觉信息来回答问题。

Text-based VQA: For the text modality, we use two evaluation formats: (1) MC-VQA: Models are given an image, a text question, and four answer options. They must select the correct option by reasoning over the image and question. (2) Open-ended VQA: Models receive an image and a text question without answer options, and are required to generate the correct answer. This tests their ability to retrieve and reason about cultural knowledge without potential hints in the answer set.

基于文本的 VQA：对于文本模态，我们使用两种评估格式：(1) 多选 VQA：模型会收到一张图片、一个文本问题以及四个答案选项。它们必须通过分析图片和问题来选择正确的选项。(2) 开放式 VQA：模型会收到一张图片和一个文本问题，而没有答案选项，需要生成正确的答案。这测试了它们在没有答案集合中潜在提示的情况下检索和推理文化知识的能力。

Audio-based VQA: Similarly, we evaluate audio modality using two formats: (1) Audio MC-VQA: Same setting as MC-VQA described above, but models are queried through African-accented English and native language speech. (2) Audio open-ended VQA: Given an image and the question in speech format without answer options and models required to generate the correct answer in text.

音频问答（Audio-based VQA）：同样，我们使用两种格式来评估音频模态：(1) 音频多项选择问答（Audio MC-VQA）：与上述描述的 MC-VQA 设置相同，但模型是通过带有非洲口音的英语和母语语音进行查询的。(2) 音频开放式问答（Audio open-ended VQA）：给定一张图片和以语音格式呈现的问题（没有答案选项），模型需要在文本中生成正确的答案。

5.2 Control Experiments

5.2 控制实验

While it is technically challenging to determine whether prediction failures on Afri-MCQA arise from limitations in language understanding or gaps in cultural knowledge, we conduct control experiments on easy tasks that primarily require language understanding in either text or speech form. These evaluations provide evidence to understand the extent to which language-understanding limitations may contribute to the observed failures.

虽然从技术上讲，确定 Afri-MCQA 上的预测失败是由于语言理解的局限性还是文化知识的差距具有挑战性，但我们进行了控制实验，这些实验主要涉及文本或语音形式的语言理解任务。这些评估提供了证据，以了解语言理解的局限性在多大程度上可能对观察到的失败有所贡献。

Text-based experiments: To probe text understanding, we evaluate on two benchmarks: (1) AfriXNLI (Adelani et al., 2025): natural language inference, and (2) AfriMMLU (Adelani et al., 2025): general knowledge QA. By analyzing performance on these text-only tasks and comparing it with results on Afri-MCQA, we obtain an approximate measure of the models' baseline linguistic competence on the studied languages.

文本实验：为了探索文本理解能力，我们在两个基准数据集上进行评估：(1) AfriXNLI（Adelani 等，2025）：自然语言推理，以及 (2) AfriMMLU（Adelani 等，2025）：通用知识问答。通过分析这些仅文本任务上的表现，并将其与 Afri-MCQA 上的结果进行比较，我们获得了一个对模型在所研究语言中基础语言能力的近似衡量。

Audio-based experiments: To probe audio understanding, we conduct two tasks: (1) ASR: transcribing spoken African language audio to text, assessing whether models can accurately capture spoken content as a prerequisite for answering questions. (2) Language Identification (LID): identifying which of the 15 languages is spoken, testing the model's ability to recognize spoken languages. These tasks reveal whether poor audio VQA results from speech-processing failures or cultural reasoning limitations.

音频相关实验：为了探索音频理解能力，我们进行了两个任务：(1) ASR：将口语化的非洲语言音频转录为文本，评估模型是否能够准确捕捉口语内容，这是回答问题的前提条件。(2) 语言识别（LID）：识别这 15 种语言中哪一种被使用，测试模型识别口语语言的能力。这些任务揭示了音频问答结果不佳的原因是由于语音处理失败还是文化推理限制。

5.3 Evaluation metrics

5.3 评估指标

We evaluate models in a zero-shot setting using automatic metrics across all tasks, with additional human evaluation for open-ended VQA (text).

我们在零样本设置下，使用自动指标对所有任务中的模型进行评估，并对开放式的 VQA（文本）进行额外的人工评估。

Automatic Evaluation: We report accuracy scores for MC-VQA and classification tasks, and use GPT-4o-mini (Hurst et al., 2024) as a judge for Open-ended VQA. For Open-ended QA, we additionally compute chrF++ (Popović, 2015) scores and present them in Appendix C. We report Word Error Rate (WER) for ASR.

自动评估：我们报告 MC-VQA 和分类任务的准确率分数，并使用 GPT-4o-mini (Hurst 等，2024) 作为开放式视觉问答的评判者。对于开放式问答，我们还计算 chrF++ (Popović, 2015) 分数，并在附录 C 中展示。我们报告语音识别的词错误率（WER）。

Human Evaluation: We evaluate 50 randomly sampled questions per language on the best-performing model per family. Bilingual native speakers rated whether model outputs matched the gold answer or were valid alternatives.

人工评估：我们在每个语言上对表现最好的模型进行评估，每种语言评估 50 个随机选取的问题。双语母语者评估模型输出是否与标准答案一致，或者是否为有效的替代答案。

6 Results

6 结果

In this Section, we present results organized by evaluation type, beginning with cultural VQA performance across MC-VQA and Open-ended VQA tasks, followed by control experiments.

在本节中，我们按评估类型组织结果，首先从在 MC-VQA 和开放式 VQA 任务上的文化 VQA 性能开始，然后是控制实验。

6.1 Cultural VQA Results

6.1 文化问答（VQA）结果

We show evaluations on MC-VQA and Open-ended VQA tasks, comparing performance across text and audio modalities in both English and native languages.

我们展示了在 MC-VQA 和开放式 VQA 任务上的评估，比较了英文和母语环境下文本和音频模态的性能。

6.1.1 Text-based QA

6.1.1 基于文本的问答

Figure 3 depicts MC-VQA average accuracy (3(a)) and LLM-as-a-judge scores (3(b)) for open-ended QA in both English and native African languages using location-aware prompts.

图 3 展示了使用位置感知提示在英语和本地非洲语言中进行开放式问答的 MC-VQA 平均准确率（3(a)）和 LLM-as-a-judge 评分（3(b)）。

Open-weight models consistently perform better when the question is in English compared to native languages across both tasks. We also observe a significant performance gap between MC-VQA and Open-ended QA, where all models, including Gemini-2.5 Pro, show major drops when tasked with answering in an open-ended setting, even in English. This suggests that generating culturally

在两个任务中，当问题为英文时，开放权重模型的表现始终优于原生语言。我们还观察到 MC-VQA 与开放式问答之间存在显著的性能差距，其中所有模型，包括 Gemini-2.5 Pro，在开放式设置下回答问题时都表现出明显的性能下降，即使在英文环境下也是如此。这表明生成文化

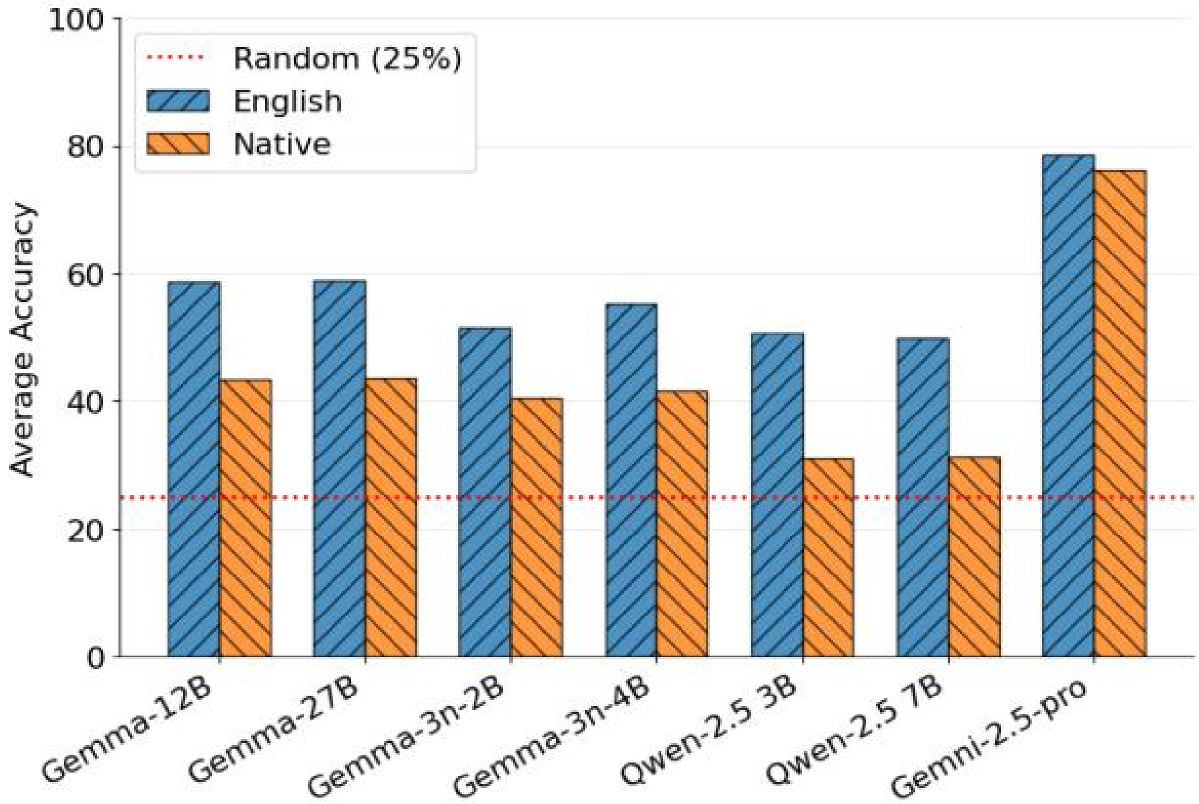

Figure 3: images/c832306eab35dbc383efb2364784c120a02585646236b6dc88485b25ad56f5ea.jpg

(a) MC-VQA (Text)

(a) MC-VQA (文本)

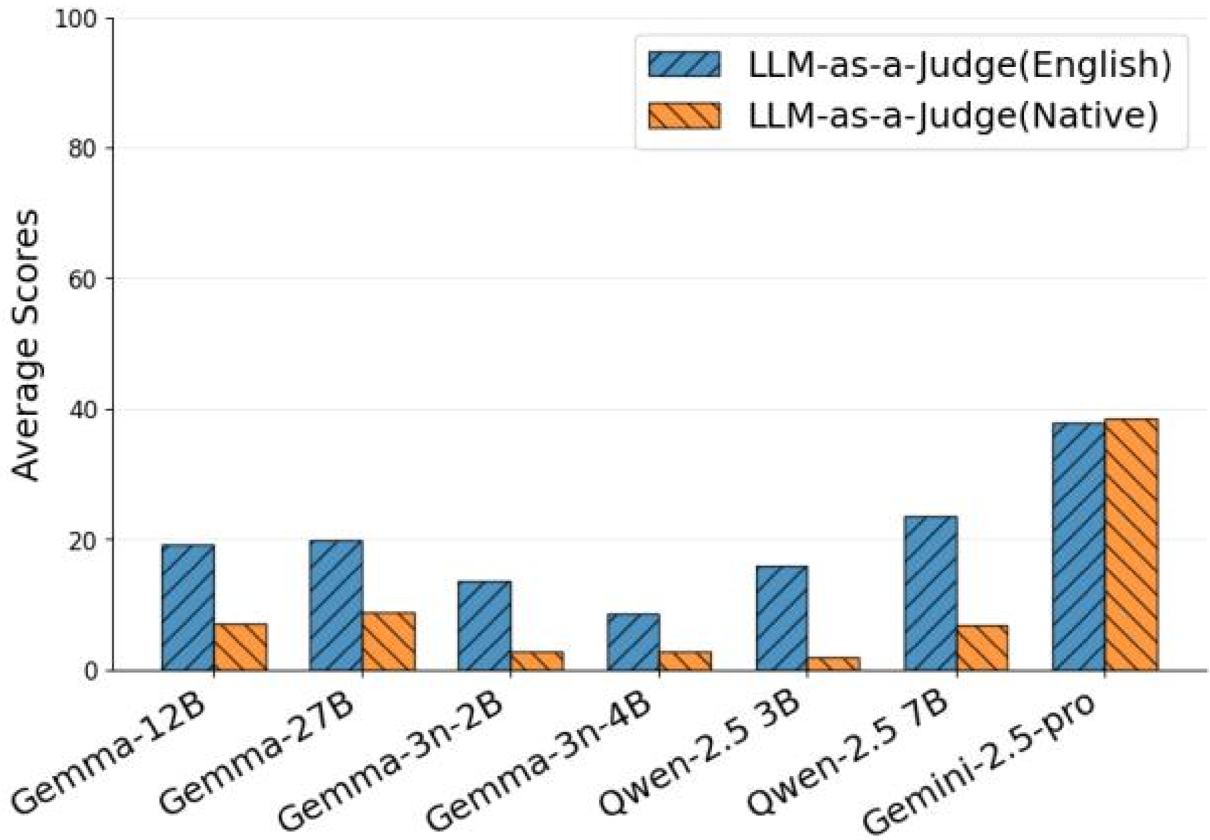

Figure 4: images/92722c17a0bfc615e7a21f427b02717d816ed08ed1ec03be63b7b293d54d7264.jpg

(b) Open-ended VQA (Text)

(b) 开放式视觉问答（文本）

Figure 3: Performance comparison of models on text-based question answering tasks: (a) Text MC-VQA (Multiple Choice) and (b) Text Open-ended QA in English and Native languages.

图 3：模型在基于文本的问答任务中的性能比较：(a) 文本多选问答（Multiple Choice）和 (b) 英文及母语的文本开放式问答。

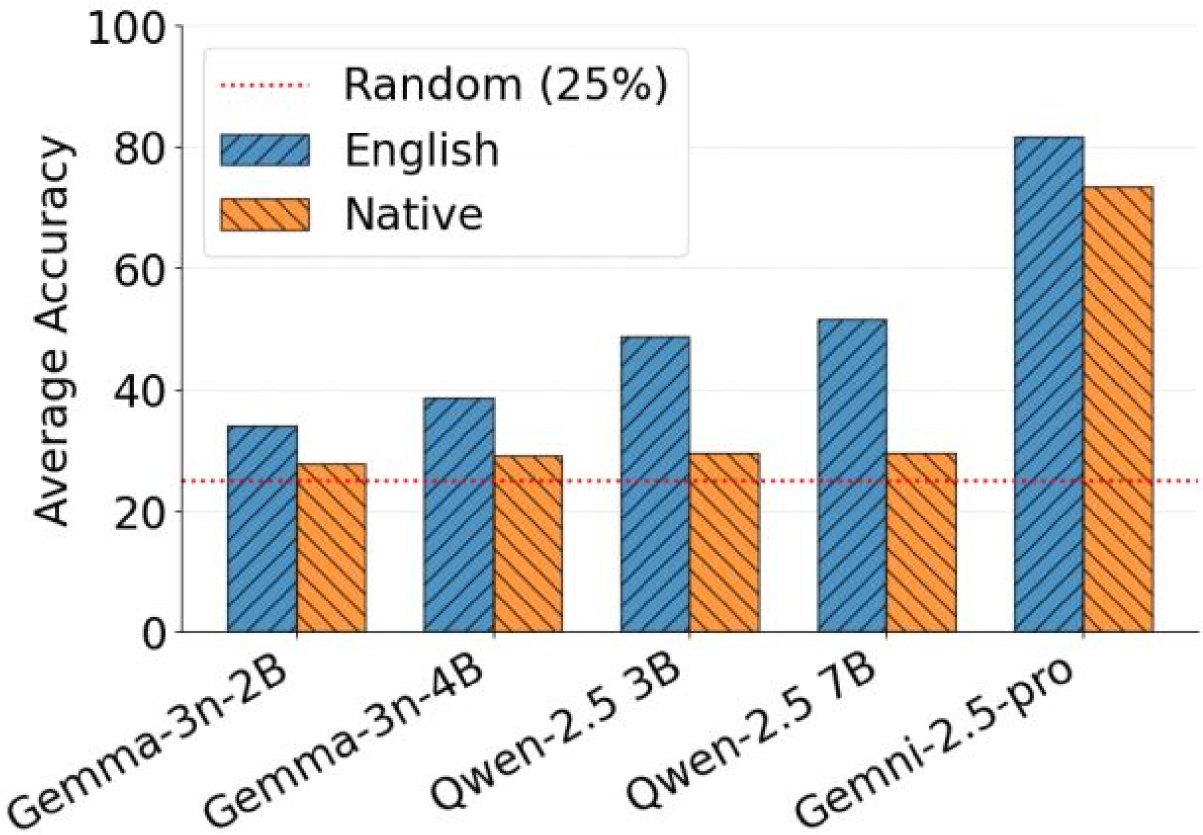

Figure 5: images/aec38b4423cf0d7ade2fe2a7820f28bed5eec21870a8c41a9571fd6eb7db5ce9.jpg

(a) MC-VQA (Audio)

(a) 基于多模态的 VQA（音频）

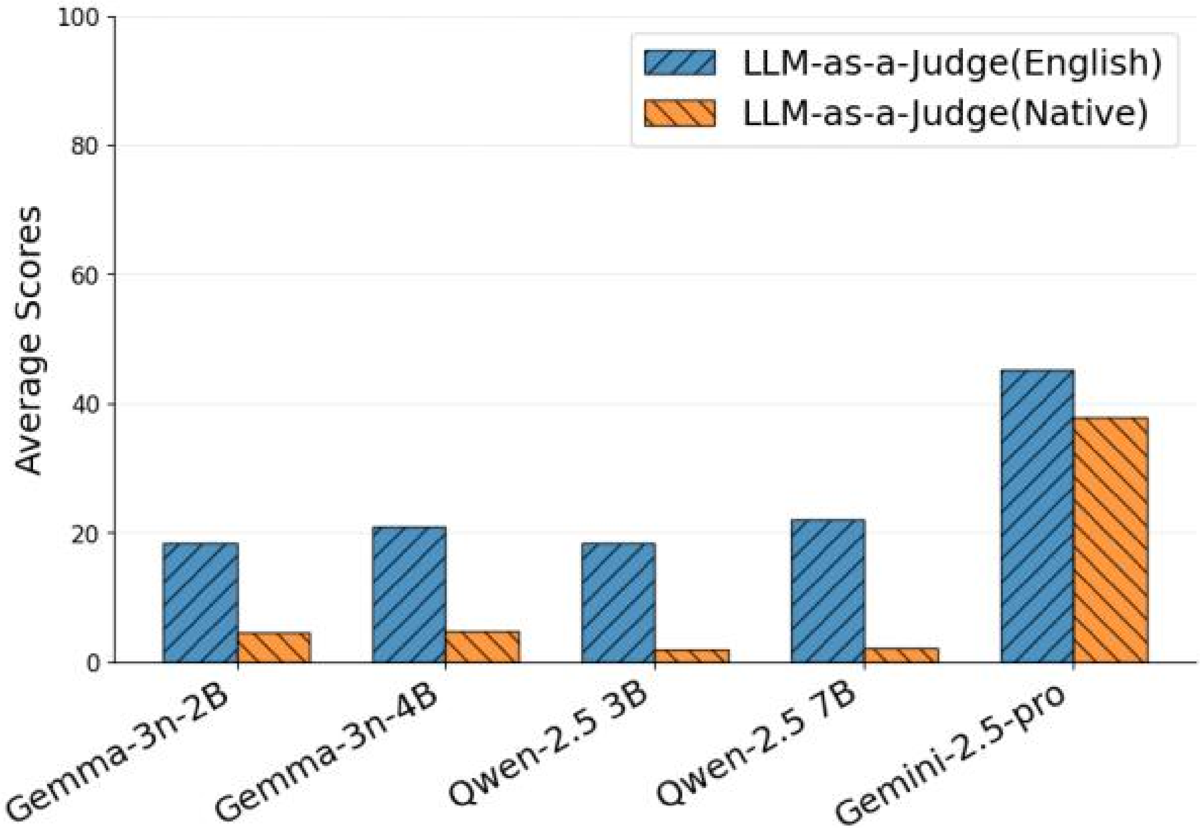

Figure 6: images/7c177abff8d65f9afb011ca1efdf3ac0e42b008aba0105c48b6f81d5486b3c72.jpg

(b) Open-ended VQA (Audio)
Figure 4: Performance comparison of models on audio-based question answering tasks: (a) Audio MC-VQA (Multiple Choice) and (b) Audio Open-ended VQA in English and Native languages.

(b) 开放式音频问答（音频）
图 4：模型在基于音频的问题回答任务中的性能比较：(a) 音频多选问答（多选）和 (b) 英语及母语的音频开放式问答。

grounded responses is more challenging than selecting from predefined options, particularly for native-language queries, where performance degrades significantly compared to English queries.

基于语境的响应比从预定义选项中选择更具挑战性，尤其是对于母语查询，其性能相比英语查询会显著下降。

We also observe that increasing model size among open-weight models does not necessarily translate to improved performance in low-resource languages. Larger variants show limited or no improvement over smaller counterparts in native language QA, indicating that model scaling alone is insufficient to address low-resource challenges. Notably, some smaller models achieve near-zero accuracy for native languages for Open-ended QA. In contrast, Gemini-2.5-Pro outperforms all open-weight models across both tasks while maintaining comparable performance between English and native languages, highlighting the current gap between proprietary and open-weight models. Human evaluations on Open-Ended VQA (in Fig ure 5), conducted on a random subset of 50 samples across eight languages, are consistent with the trends observed in automatic evaluations, with Gemini-2.5 Pro performs the best when queried in native languages compared to English.

我们还观察到，在开放权重模型中，模型规模的增加并不一定意味着在低资源语言上的性能提升。较大的

变体在母语问答任务中相对于较小的模型仅表现出有限或没有改进，这表明仅依靠模型扩展无法解决低资源语言的挑战。值得注意的是，一些较小的模型在开放性问答任务中对母语的准确率接近于零。相比之下，Gemini-2.5-Pro 在两项任务中均优于所有开放权重模型，同时在英语和母语之间保持相当的性能，这突显了专有模型与开放权重模型之间的当前差距。在图 5 中，对开放性视觉问答（Open-Ended VQA）进行的人类评估，基于八个语言中 50 个随机样本的子集，与自动评估中观察到的趋势一致，当使用母语进行查询时，Gemini-2.5-Pro 的表现优于英语。

6.1.2 Audio-based QA

6.1.2 基于音频的问答

Figure 4 shows MC-VQA accuracy (4(a)) and LLM-as-a-judge scores for Open-ended QA (4(b)) on audio inputs in both English and native African languages. Similar to text-based evaluation, open-weight models achieve higher performance when queried in English compared to native languages across both tasks. For open-weight models, audio modality is significantly more difficult than text modality, with notable performance degradation across both tasks. Gemini-2.5-Pro, however, demonstrates robust multimodal capabilities, with consistent performance across modalities. The

图 4 展示了在英语和本土非洲语言的音频输入上，MC-VQA 准确率（4(a)）和开放式问答（4(b)）中 LLM 作为评判者的得分。与基于文本的评估类似，在两个任务中，当使用英语进行查询时，开放权重模型的表现优于本土语言。对于开放权重模型而言，音频模态比文本模态要困难得多，在两个任务中都出现了显著的性能下降。然而，Gemini-2.5-Pro 展示了强大的多模态能力，在不同模态下表现一致。

performance drop from MC-VQA to Open-ended QA is also noticeable in the audio modality, with nearly zero accuracy in spoken native languages. These results align with our LID and ASR analyses (Section 6.2.2). Open-weight models demonstrate poor LID capabilities, especially the Qwen variants, which exhibit near-random accuracy and significantly compromised native ASR performance compared to English. These failures also affect performance models in downstream tasks, such as Open-ended VQA. Similar to text evaluations, we find that scaling up model size shows little improvement in native language understanding.

从 MC-VQA 到开放式问答的性能下降在音频模态中也很明显，特别是在口语母语中几乎准确率为零。这些结果与我们的语言识别（LID）和自动语音识别（ASR）分析结果一致（第 6.2.2 节）。开放权重模型在 LID 方面表现不佳，尤其是 Qwen 的变体，其准确率接近随机，且在母语 ASR 性能上相比英语有显著下降。这些失败也影响了下游任务中的性能模型，例如开放式的视频问答（VQA）。与文本评估类似，我们发现扩大模型规模对母语理解的提升效果很小。

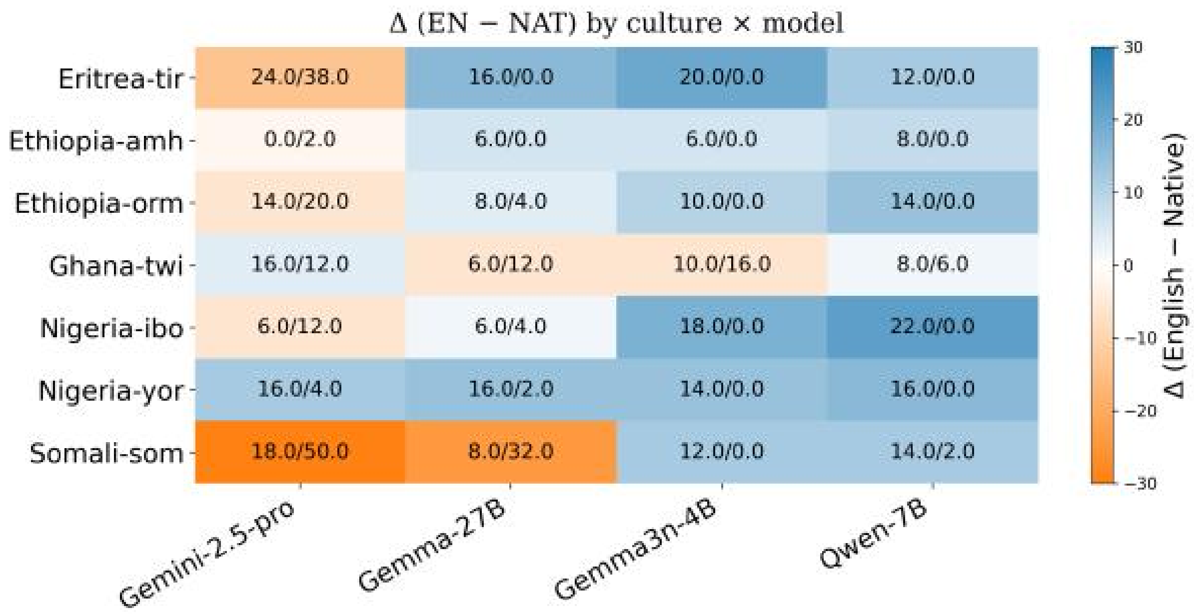

Figure 7: images/845200d1efe2fecba409c4d215ef69669623c888c9e985a6c2d5e0d2baa3daca.jpg

Figure 5: Human Evaluation for Text Open-ended VQA. Accuracy across best-performing models for English and Native. We observed that, while most models perform best in the English setting, Gemini-2.5 Pro seems to perform better in the Native language.

图 5：文本开放式 VQA 的人类评估。英语和母语环境下表现最好的模型的准确性。我们观察到，虽然大多数模型在英语环境下表现最佳，但 Gemini-2.5 Pro 似乎在母语环境下表现更好。

6.2 Results for Control Experiments

6.2 控制实验结果

We present the results on control experiments on established benchmarks alongside our cultural QA task, aiming to probe the linguistic competence of the tested models and provide pointers on why models fail on Afri-MCQA.

我们展示了在已建立的基准测试中进行控制实验的结果，以及我们的文化问答任务，旨在探究被测试模型的语言能力，并提供模型在 Afri-MCQA 上失败的原因。

6.2.1 Text-based Experiments

6.2.1 基于文本的实验

In Table 3, we report the performance of the models on the control-experiment benchmarks in addition to Afri-MCQA. All models showed performance degradation from English to native languages across all benchmarks. Gemini-2.5-Pro maintains the smallest gaps, particularly on Afri-MCQA, where the drop is minimal. Open-weight models show a big drop, with Qwen variants showing the most severe gaps on AfriXNLI and AfriMMLU.

在表 3 中，我们报告了模型在控制实验基准测试中的表现，除了 Afri-MCQA 之外。所有模型在所有基准测试中从英语到母语的表现都有所下降。Gemini-2.5-Pro 保持了最小的差距，特别是在 Afri-MCQA 上，

其下降幅度最小。开放权重模型表现出较大的下降，其中 Qwen 变体在 AfriXNLI 和 AfriMMLU 上表现出最严重的差距。

When comparing AfriMMLU with Afri-MCQA, open-weight models show considerably higher AfriMMLU scores in English than Afri-MCQA scores, suggesting that while models possess general factual knowledge, they lack Africa-specific cultural understanding. Gemini-2.5-Pro shows a smaller gap between these benchmarks, indicating it has acquired more African cultural knowledge during training. AfriXNLI exposes the most severe cross-lingual gaps, particularly for Qwen models, suggesting that linguistic reasoning tasks are more sensitive to language resource availability than factual retrieval tasks.

当将 AfriMMLU 与 Afri-MCQA 进行比较时，开放权重模型在英语中的 AfriMMLU 得分显著高于 Afri-MCQA 得分，这表明虽然模型具备一般性的事实知识，但缺乏针对非洲特定的文化理解。Gemini-2.5-Pro 在这两个基准之间的差距较小，表明它在训练过程中获得了更多的非洲文化知识。AfriXNLI 暴露出最严重的跨语言差距，尤其是对于 Qwen 模型而言，这表明语言推理任务比事实检索任务更敏感于语言资源的可用性。

We additionally compute Spearman rank correlations between Afri-MCQA and our text control datasets (see Appendix D). We observe strong and statistically significant correlations between AfriXNLI and AfriMMLU performance for several models. In contrast, correlations between Afri-MCQA and AfriXNLI or AfriMMLU are generally weaker and not statistically significant. Hence, while limitations in language understanding may play a large role in explaining why models fail in Afri-MCQA, other factors related to African cultural and visual knowledge may be important as well.

我们还计算了 Afri-MCQA 与我们的文本控制数据集之间的斯皮尔曼等级相关性（见附录 D）。我们观察到，对于多个模型而言，AfriXNLI 与 AfriMMLU 的表现之间存在强且统计上显著的相关性。相比之下，Afri-MCQA 与 AfriXNLI 或 AfriMMLU 之间的相关性通常较弱且不具有统计显著性。因此，虽然语言理解能力的限制可能在解释模型为何在 Afri-MCQA 中表现不佳方面起着重要作用，但与非洲文化及视觉知识相关的其他因素也可能是重要的。

Model	AfriXNLI			AfriMMLU			Afri-MCQA		
Eng	Nat	Δ	Eng	Nat	Δ	Eng	Nat	Δ	
Gemini-2.5-Pro	89.66	76.3	-13.36	94	83.46	-10.54	78.68	76.27	-2.4
Gemma3n-4B	78.33	51.24	-27.09	62.2	37.56	-24.64	55.23	41.49	-13.7
Gemma3n-2B	80.66	51.47	-29.19	53.6	35.74	-17.86	51.54	40.32	-11.2
Qwen-7B	81.11	34.33	-46.78	73.8	36.10	-37.70	49.75	31.30	-18.46
Qwen-3B	65.66	36.7	-28.96	65.8	34.72	-31.08	50.68	31.00	-19.65

模型	AfriXNLI			AfriMMLU			Afri-MCQA		
英	Nat	Δ	英	Nat	Δ	英	Nat	Δ	
Gemini-2.5-Pro	89.66	76.3	-13.36	94	83.46	-10.54	78.68	76.27	-2.4
Gemma3n-4B	78.33	51.24	-27.09	62.2	37.56	-24.64	55.23	41.49	-13.7
Gemma3n-2B	80.66	51.47	-29.19	53.6	35.74	-17.86	51.54	40.32	-11.2
Qwen-7B	81.11	34.33	-46.78	73.8	36.10	-37.70	49.75	31.30	-18.46
Qwen-3B	65.66	36.7	-28.96	65.8	34.72	-31.08	50.68	31.00	-19.65

Table 3: Text-based performance (%). Eng = English accuracy, Nat = Native accuracy, Δ = English – Native gap (negative = drop).

表 3：基于文本的性能（%）。Eng = 英语准确率，Nat = 母语准确率，Δ = 英语 – 母语差距（负值 = 下降）。

6.2.2 Audio-based Experiments

6.2.2 基于音频的实验

Figure 6 shows three evaluations on native African language audio. LID: Gemini-2.5-Pro achieves near-perfect LID accuracy, while Gemma variants show moderate performance. Qwen models perform at near random levels, indicating minimal exposure to African language audio during pretraining. ASR Performance: WER shows a similar pattern. Gemini-2.5-Pro maintains reasonable native ASR, whereas Gemma models exhibit substantial degradation. Qwen models produce high error rates, indicating hallucinations rather than meaningful transcription. This aligns with observed results for open-ended VQA (Audio). Gemini-2.5-Pro achieves moderate accuracy, whereas open-weight models score near zero. In light of these results, it is likely that models fail on open-ended VQA (Audio) because they cannot properly identify or transcribe the tested spoken languages.

图 6 展示了对原生非洲语言音频的三项评估。语言识别（LID）：Gemini-2.5-Pro 实现了接近完美的 LID 准确率，而 Gemma 变体的表现中等。Qwen 模型的表现接近随机，表明其在预训练过程中对非洲语言音频的接触非常有限。语音识别（ASR）表现：WER 呈现出类似模式。Gemini-2.5-Pro 保持了合理的原生 ASR 表现，而 Gemma 模型则表现出显著的性能下降。Qwen 模型产生高错误率，表明其产生的是幻觉而非有意义的转录。这与开放性视觉问答（音频）的观察结果一致。Gemini-2.5-Pro 实现了中等准确率，而开放权重模型的得分接近于零。鉴于这些结果，模型在开放性视觉问答（音频）任务上表现不佳的原因很可能是它们无法正确识别或转录所测试的口语语言。

Hence, poor Open-ended VQA (Audio) results come from errors at each step: (1) Open models often fail at identifying African languages, reflecting fundamental gaps in audio representation, (2)

因此，开放式的音频问答（Open-ended VQA (Audio)）结果不佳源于每一步的错误：(1) 开放模型在识别非洲语言时常常失败，反映了音频表示中的根本性差距，(2)

they show high ASR errors, suggesting that most spoken content is not perceived adequately for understanding and answering a question (3) these errors can carry over to open-ended VQA when models receive wrong transcriptions, as they cannot answer correctly even if they know the cultural content. These findings demonstrate that foundational speech processing capabilities are prerequisites for meaningful evaluation of cultural reasoning in African languages.

它们显示出较高的语音识别错误率，表明大部分口语内容未能被充分理解，从而无法回答问题。(3) 当模型收到错误的转录时，这些错误可能会延续到开放式的视觉问答任务中，即使它们了解文化内容，也无法正确回答问题。这些发现表明，基础的语音处理能力是评估非洲语言中文化推理能力的前提条件。

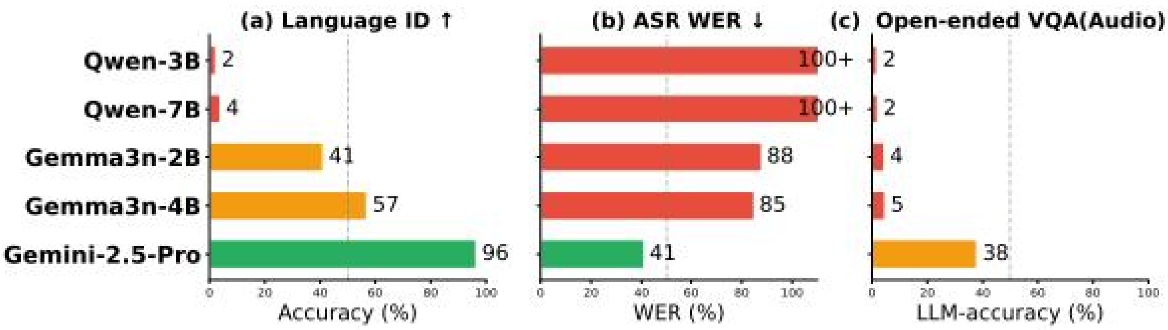

Figure 8: images/9c997b55bc773d2911a3f37bf7419ad2866a4bc30ae42f3609162b5859cf9c25.jpg

Figure 6: Audio probing results on native African languages: (a) LID (↑ higher is better), (b) ASR (↓ lower is better), and (c) Open-ended VQA (↑ higher is better). (+) means WER is more than

100%

图 6：在本土非洲语言上的音频探测结果：(a) 语言识别（LID）（↑ 数值越高越好），(b) 语音识别（ASR）（↓ 数值越低越好），以及 (c) 开放式视觉问答（Open-ended VQA）（↑ 数值越高越好）。(+) 表示 WER 超过 100%

7 Discussion

7 讨论

We now organize our findings to address each research question, summarizing key patterns observed across models, modalities, and languages.

我们现在整理我们的发现，以回答每个研究问题，总结在模型、模态和语言中观察到的关键模式。

RQ1: How well do MLLMs understand African cultural contexts in visually-grounded QA? MLLMs show limited understanding of African cultural contexts. As shown in Figures 3 and 4, even the best performing model (Gemini-2.5-Pro) achieves only 78% on MC-VQA and 38% on Open-VQA (text-based, English). Smaller models (Gemma3, Qwen2.5) perform substantially worse, ranging from 50–59% on MC-VQA and 9–24% on Open-VQA, indicating significant room for improvement.

RQ1: MLLMs 在基于视觉的问答任务中对非洲文化背景的理解程度如何？MLLMs 对非洲文化背景的理解有限。如图 3 和图 4 所示，即使表现最好的模型（Gemini-2.5-Pro）在 MC-VQA 中也仅达到 78%，在 Open-VQA（基于文本，英文）中也仅达到 38%。较小的模型（Gemma3、Qwen2.5）表现明显更差，在 MC-VQA 中得分范围为 50–59%，在 Open-VQA 中得分范围为 9–24%，表明仍有显著的改进空间。

RQ2: How does input modality (text vs. speech) affect performance? Performance degrades when switching from text to speech input, particularly for smaller models. As shown in Figures 3 and 4, on MC-VQA, Qwen models drop by approximately 1–2% while Gemma models show mixed results. The gap is more visible in Open-VQA, where audio-based native language queries yield near-zero accuracy (2–5%) for smaller models. Control experiments for audio show that this comes from poor language identification (2–4% for Qwen) and high ASR word error rates (85–100%+ for non-Gemini models).

RQ2: 输入模态（文本与语音）如何影响性能？从文本切换到语音输入时，性能会下降，尤其是在较小的模型中。如图 3 和图 4 所示，在 MC-VQA 任务中，Qwen 模型的性能下降约 1–2%，而 Gemma 模型的表现则参差不齐。在 Open-VQA 任务中，这种差距更加明显，其中基于音频的本族语言查询在较小模型中几乎达到零准确率（2–5%）。音频控制实验表明，这种性能下降来自于语言识别能力差（Qwen 为 2–4%）和高语音识别（ASR）词错误率（非 Gemini 模型为 85–100%+）。

RQ3: How does query language (native vs. English) affect performance, and do differences re-

RQ3: 查询语言（原生语言 vs. 英语）如何影响性能，差异是否会影响结果？

flect language understanding or cultural knowledge gaps? English queries consistently outperform native language queries across all models and settings. As shown in Figure 3, the gap ranges from 2% (Gemini-2.5-Pro on MC-VQA) to 19% (Qwen on MC-VQA). Control experiments on AfriXNLI and AfriMMLU (Table 3) show that language understanding gaps ($\Delta = 13–47\%$) are substantially larger than cultural knowledge gaps ($\Delta = 2–19\%$ on Afri-MCQA), suggesting language understanding is the dominant limitation. However, models also struggle with cultural QA in English, indicating that both linguistic and cultural limitations contribute to poor performance.

语言理解或文化知识的差距？英语查询在所有模型和设置中均优于原语言查询。如图 3 所示，差距范围从 2%（Gemini-2.5-Pro 在 MC-VQA 上的表现）到 19%（Qwen 在 MC-VQA 上的表现）。在 AfriXNLI 和 AfriMMLU 上的控制实验（表 3）表明，语言理解差距（$\Delta = 13–47\%$）明显大于文化知识差距（$\Delta = 2–19\%$ 在 Afri-MCQA 上的表现），这表明语言理解是主要的限制因素。然而，模型在英语文化问

答任务上也存在困难，表明语言和文化方面的限制共同导致了表现不佳。

RQ4: How does task format (Multiple-Choice vs. Open QA) affect accuracy? As shown in Figures 3 and 4, models perform significantly better on MC-VQA than Open-VQA. Gemini-2.5-Pro achieves 78% on MC-VQA vs. 38% on Open-VQA (a 40% gap). Smaller models show even larger relative drops, with some scoring less than 10% on Open-VQA. This performance gap suggests that MC-VQA benefits from simplified answer selection, while Open-VQA exposes true limitations. Models struggle to generate culturally grounded responses even in English, with performance degrading further for native languages.

RQ4: 任务格式（多项选择与开放问答）如何影响准确性？如图 3 和图 4 所示，模型在 MC-VQA 上的表现显著优于 Open-VQA。Gemini-2.5-Pro 在 MC-VQA 上达到 78%，而在 Open-VQA 上仅为 38%（差距达 40%）。较小的模型表现出更大的相对下降，有些模型在 Open-VQA 上的得分甚至低于 10%。这种性能差距表明，MC-VQA 得益于简化答案选择，而 Open-VQA 则暴露了模型的真实局限性。即使在英语中，模型也难以生成具有文化背景的回答，对于母语的回答表现更是进一步下降。

8 Conclusion

8 结论

We introduced Afri-MCQA, the first large-scale multilingual and multimodal benchmark for African cultural visual QA, covering 15 languages across 12 countries with over 7.5k Q&A pairs in text and speech modalities. Our evaluation shows: 1) MLLMs struggle significantly with African cultural knowledge, 2) speech processing presents a critical bottleneck, and 3) gaps persist across languages and task formats.

我们介绍了 Afri-MCQA，这是首个针对非洲文化视觉问答的大型多语言和多模态基准数据集，涵盖 12 个国家的 15 种语言，包含超过 7500 个文本和语音模态的问答对。我们的评估结果表明：1）大语言模型在非洲文化知识方面存在显著困难；2）语音处理存在关键瓶颈；3）不同语言和任务格式之间仍存在差距。

These findings motivate several research directions: (1) speech-first approaches: Many African languages are primarily oral, yet current open-weight models lack basic LID and ASR capabilities for these languages; (2) culturally-grounded pretraining: The gap between AfriMMLU and Afri-MCQA performance suggests language data alone is insufficient; models need explicit exposure to African cultural content; and (3) cross-lingual cultural transfer: Models may “know” cultural facts in English but cannot access them through native language queries, motivating research into cross-lingual knowledge retrieval.

这些发现激发了几个研究方向：(1) 以语音为主的方案：许多非洲语言主要是口头语言，但目前的开放权重模型缺乏对这些语言的基本语言识别（LID）和语音识别（ASR）能力；(2) 基于文化的预训练：AfriMMLU 与 Afri-MCQA 表现之间的差距表明，仅靠语言数据是不够的；模型需要明确接触非洲文化内容；(3) 跨语言文化迁移：模型可能 “知道” 英语中的文化事实，但无法通过母语查询访问这些信息，这促使人们研究跨语言知识检索。

We release Afri-MCQA to provide both a benchmark and a foundation for building more inclusive, culturally aware multimodal systems that better represent African languages and cultures.

我们发布 Afri-MCQA，旨在提供一个基准，并为构建更具包容性、文化意识的多模态系统奠定基础，这些系统能够更好地代表非洲语言和文化。

9 Limitations

9 局限性

We believe Afri-MCQA represents an important step toward more inclusive evaluation by foregrounding African languages and cultural contexts that have long been overlooked in existing benchmarks. Although the dataset spans 15 languages across 12 countries, Africa is home to thousands of languages and cultural groups, many of which remain unrepresented. Furthermore, while our question categories aim to reflect culturally grounded knowledge, culture itself is fluid, subjective, and deeply contextual. Our formulation inevitably abstracts away from finer-grained variations such as regional, generational, or community-specific differences that shape cultural understanding. As with most human-curated datasets, potential biases in data collection remain. Annotators' backgrounds and interpretations of 'cultural relevance' may influence the formulation of questions or the selection of images. Additionally, due to computational and financial constraints, we evaluate only a limited set of open- and closed-source models, so the reported performance gaps may not fully capture the broader landscape. Finally, Afri-MCQA is a human-curated dataset created without the involvement of LLMs and with minimal reliance on web-sourced images. Because this process is inherently time-consuming, the dataset is of moderate size and intended as an evaluation benchmark rather than a pretraining or fine-tuning resource, for which it would likely cause overfitting.

我们认为 Afri-MCQA 代表了朝着更具包容性的评估迈出的重要一步，因为它突出了长期以来在现有基准中被忽视的非洲语言和文化背景。尽管该数据集涵盖 15 种语言，涉及 12 个国家，但非洲拥有数千种语言和文化群体，其中许多仍未得到代表。此外，虽然我们的问题类别旨在反映基于文化的知识，但文化本身是流动的、主观的，并且深深植根于具体情境。我们的表述不可避免地忽略了诸如地区性、代际差异或社区特定差异等更细粒度的变化，这些变化塑造了文化理解。与大多数人工整理的数据集一样，数据收集过程中仍可能存在潜在的偏见。标注者的背景以及对 “文化相关性” 的理解可能会影响问题的制定或图像的选择。此外，由于计算和财务限制，我们仅评估了有限数量的开源和闭源模型，因此报告的性能差异可能无法完全反映更广泛的状况。最后，Afri-MCQA 是一个在没有涉及大语言模型（LLMs）且仅少量依赖网络来源图像的情况下人工整理的数据集。由于这一过程本质上耗时，该数据集规模适中，旨在作为评估基准，而不是用于预训练或微调的资源，因为使用它可能会导致过拟合。

10 Ethical Considerations

10 伦理考量

Our work involves the collection of culturally grounded question–answer pairs in 15 African languages, annotated, spoken and reviewed by native speakers. All annotators participated voluntarily and were compensated fairly for their work in accordance with local wage standards on the Upwork platform. Before beginning annotation, contributors were informed about the goals of the project, the intended use of the dataset for research and evaluation purposes, and their right to withdraw from participation at any stage.

我们的工作涉及在 15 种非洲语言中收集具有文化背景的问题-答案对，这些内容由母语者进行标注、口语化处理和审核。所有标注人员都是自愿参与的，并且根据 Upwork 平台上的当地工资标准，公平地获得了其工作的报酬。在开始标注之前，参与者被告知了项目的宗旨、数据集用于研究和评估的目的，以及他们在任何阶段都有权退出参与的权利。

We took several steps to ensure cultural sensitivity and respect throughout the data creation process.

我们采取了多项措施，以确保在整个数据创建过程中体现出文化敏感性和尊重。

Question formulation guidelines were designed to avoid harmful stereotypes, offensive content, or culturally inappropriate framing. All annotations were reviewed by language coordinators who are themselves native speakers to check for accuracy, contextual appropriateness, and respectful representation. Despite these efforts, we acknowledge that culture is deeply complex and subjective, and that our dataset may still reflect certain biases or oversimplifications.

问题构建指南的设计旨在避免有害的刻板印象、冒犯性内容或文化不合适的表述。所有注释均由语言协调员进行审核，他们本身就是母语者，以检查准确性、语境适宜性以及尊重性的呈现。尽管我们做出了这些努力，我们承认文化是深具复杂性和主观性的，我们的数据集可能仍然反映出某些偏见或过度简化。

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